

LA HW

Included exercises: 7, 15, 81-89, 92, 95

Exercises 1-8

In Exercises 1–8, two vectors $\mathbf{u}$ and $\mathbf{v}$ are given. Compute the norms of the vectors and the distance d between them.

Original Exercise 7

$$\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ -2 \\ 1 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ 1 \\ 1 \end{bmatrix}$$

Exercises 9-16

In Exercises 9–16, two vectors are given. Compute the dot product of the vectors, and determine whether the vectors are orthogonal.

Original Exercise 15

$$\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ -2 \\ 1 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ 1 \\ 1 \end{bmatrix}$$

Original Exercise 81

Prove (a) of Theorem 6.1.

Original Exercise 82

Prove (b) of Theorem 6.1.

Original Exercise 83

Prove (c) of Theorem 6.1.

Original Exercise 84

Prove (e) of Theorem 6.1.

Original Exercise 85

Prove (f) of Theorem 6.1.

Original Exercise 86

Prove that if $\mathbf{u}$ is orthogonal to both $\mathbf{v}$ and $\mathbf{w}$, then $\mathbf{u}$ is

Original Exercise 87

Let $\{\mathbf{v}, \mathbf{w}\}$ be a basis for a subspace W of $\mathcal{R}^n$, and define

$$\mathbf{z} = \mathbf{w} - \frac{\mathbf{v} \cdot \mathbf{w}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

Prove that $\{\mathbf{v}, \mathbf{z}\}$ is a basis for W consisting of orthogonal

Exercise 88

124. Use Exercise 88 to prove that the two averaging methods for determining class size are equal if and only if all of the class sizes are equal.

Original Exercise 88

vectors.

Prove that the Cauchy–Schwarz inequality is an equality if and only if $\mathbf{u}$ is a multiple of $\mathbf{v}$ or $\mathbf{v}$ is a multiple of $\mathbf{u}$.

Original Exercise 89

Prove that the triangle inequality is an equality if and only if $\mathbf{u}$ is a nonnegative multiple of $\mathbf{v}$ or $\mathbf{v}$ is a nonnegative

Original Exercise 92

Let $\mathbf{z}$ be a vector in $\mathcal{R}^n$. Let $W = \{\mathbf{u} \in \mathcal{R}^n : \mathbf{u} \cdot \mathbf{z} = 0\}$.

Original Exercise 95

$W = \{\mathbf{u} \in \mathcal{R}^2 : \mathbf{u} \cdot \mathbf{z} = 0\}$. Justify your answer.

Prove the *parallelogram law* for vectors in $\mathcal{R}^n$:

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2.$$